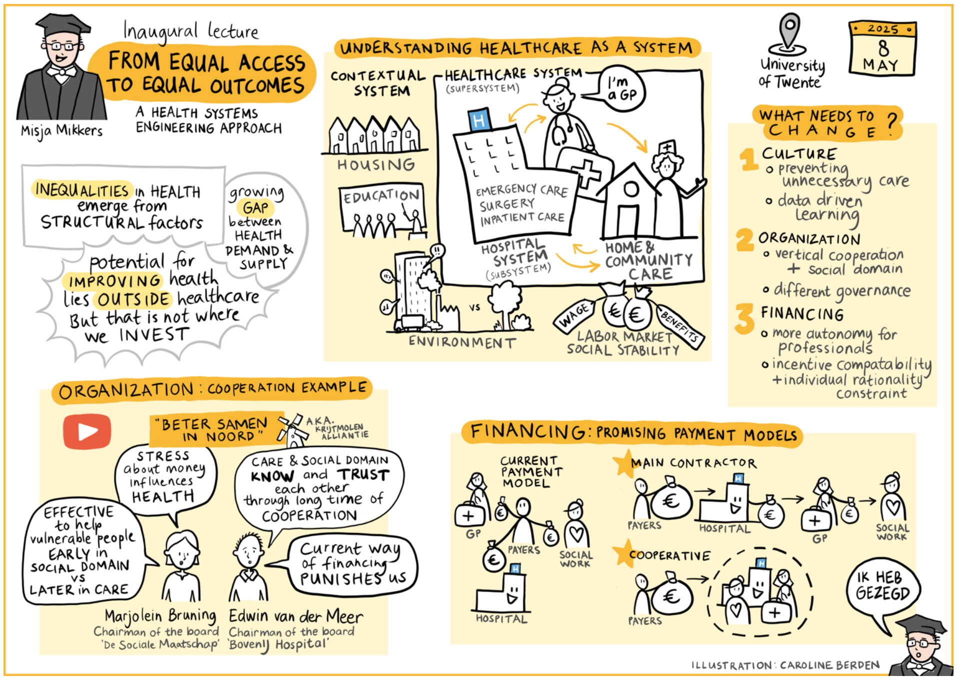

From Equal Access to Equal Outcomes

Inaugural Address University of Twente

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May 8, 2025

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Preface

All Western countries face a growing challenge in healthcare: demand is rising due to an ageing population and ongoing technological innovation, while at the same time, the supply of healthcare workers is shrinking. This imbalance is not temporary or isolated, it seems to be structural and systematic. It forces us to rethink how we organize care, how we allocate resources, and what we define as good policy.

At the same time, this challenge presents an opportunity, an opportunity to shift focus from equal access to healthcare to equal health outcomes. Research consistently shows that the main drivers of health lie outside the healthcare sector: poverty, education, housing, and social conditions all lead to chronic diseases and health inequality. The solution is not to spend ever more on healthcare, but to invest in what actually improves outcomes. Better results come not from doing more of the same, but from doing different things and doing the right things. That means rethinking how we collaborate across sectors, how we align incentives, and how we enable systems to learn and adapt.

This text builds on my inaugural lecture at the University of Twente, where I argued that many healthcare reforms fail not due to a lack of effort or insight, but because they treat problems in isolation. Fragmented responsibilities, poorly aligned incentives, and non-existing or slow learning cycles make it difficult to provide coordinated, patient-centered care. Addressing these challenges requires a shift in perspective.

Health Systems Engineering offers such a perspective. Drawing on contract theory, systems thinking, and institutional economics, it provides a framework to understand how governance structures, payment models, and cultural factors interact. The goal is to design systems that stimulate coordination, align incentives with what matters, and stimulate learning under uncertainty.

This text goes beyond the lecture. It offers a more systematic and detailed treatment of the concepts, mechanisms, and international examples. Its central focus is on payment models, not because payments are the only thing that matters, but because they show how deeply our systems are influenced by incentives, rules, and organizational design. By analyzing these mechanisms in depth, I argue for the need of a more holistic view, one that connects financial structures to broader questions of governance and system design.

The text concludes by looking ahead. What would a better system look like? What trade-offs are inevitable? And how can we experiment and learn our way towards more sustainable, fair, and effective healthcare systems?

These are not just technical questions. They are political questions as well, about how we define public values, how we structure accountability, and how we govern collective resources in the face of demographic, social, and technological change.

I hope this text contributes to that conversation in classrooms, research projects, policy discussions, and collaborative experiments in the field.

I would like to thank Peter Bogetoft, Victora Shestalova, Merlijn Mikkers, Sandra Kompier and Finn Mikkers for their valuable comments on the text.

The cover illustration was created by Caroline Berden, to whom I am grateful for her contribution to the visual identity of this book.

1 Introduction

In many countries governments, policymakers, and researchers are concerned about the rising cost of healthcare. Health spending continues to grow faster than economic growth in most countries, and this trend is projected to continue as populations age and medical innovation accelerates. The European Commission, OECD, WHO, and national advisory bodies such as the WRR in the Netherlands have all expressed concern over the long-term sustainability of healthcare systems (European Union (2021) , OECD (2023), Visser et al. (2021)).

Yet, rising costs are not necessarily a sign of failure. Much of this spending reflects medical progress. Advances in treatment and technology have led to significantly improved survival rates for conditions such as cancer and cardiovascular disease (OECD (2023), David M. Cutler, Rosen, and Vijan (2006)).

It is important, however, to assess not just how much we spend, but also how we allocate this spending. As David M. Cutler, Rosen, and Vijan (2006) observed, while increased spending since the 1960s has often provided reasonable value, some areas, such as residential elderly care since the 1980s have seen more modest gains at increasingly higher cost. This underlines the importance of considering not only the volume of spending, but whether resources are allocated in ways that produce the best outcomes.

In my inaugural address at Tilburg University (Mikkers (2016)), I argued that despite differences in institutional design, healthcare systems in Western countries are similar in many ways, not only in the challenges they face, such as rising costs, aging populations, and workforce pressure, but also in their overall performance. Countries with very different financing models, such as tax-funded systems and social insurance schemes, tend to show comparable trends in health spending and similar gains in life expectancy and survival.

To support this claim, I estimated a Bayesian multilevel regression model using health expenditure data from 29 OECD countries, available through the OECD Data Explorer. The model included country-specific intercepts and slopes for health spending as a share of GDP, with year (centered) as a continuous predictor. This structure allowed us to estimate both a global trend and country-level deviations. A Bayesian approach was chosen for its natural fit with hierarchical data, full uncertainty quantification, and interpretable 95% credible intervals.

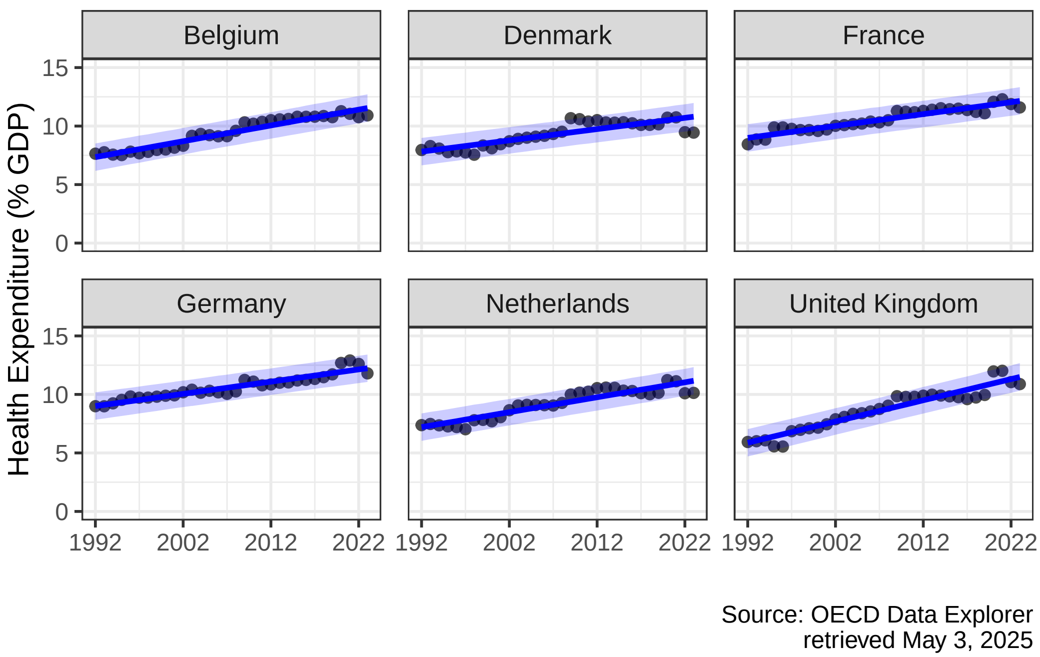
To illustrate these findings, I present results for six countries that are geographically and economically close to the Netherlands, but differ in healthcare system type:

- Germany and Belgium operate Bismarckian social health insurance systems with uniform benefits and centrally negotiated prices.

- France combines public insurance with a mix of public and private provision.
- Denmark and the United Kingdom have Beveridge-type, tax-funded systems with strong public provision.
- The Netherlands, while originally Bismarckian, introduced regulated competition in 2005.

Despite institutional differences, the estimated trends in health expenditure were remarkably similar. The global average increase in health spending was estimated at 0.10 percentage points per year (95% CrI: 0.08, 0.12). Country-specific trends were clustered closely around this global mean: the Netherlands (0.128), Belgium (0.135), Germany (0.104), France (0.102), and Denmark (0.096). The United Kingdom was the exception, with a steeper trend of 0.181 (95% CrI: 0.152, 0.211). These results suggest that despite different organizational models, long-term health spending trajectories, and by implication, some aspects of performance, are quite similar across countries.

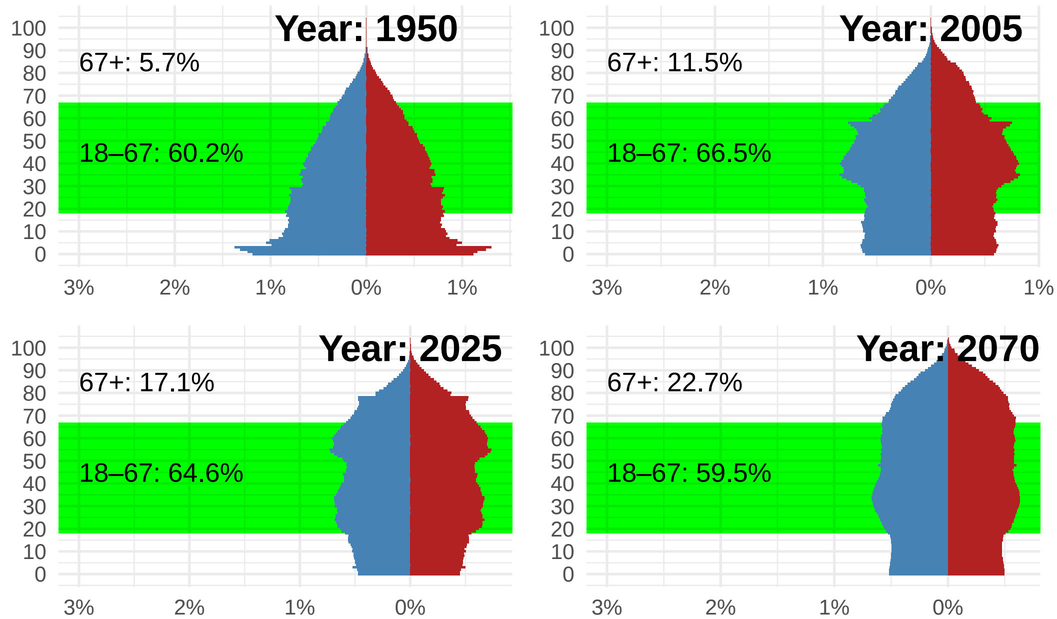
These findings are visualized in the figure below, where the observed and predicted health expenditures (with 95% credible intervals) are plotted for each country.



Yet the financial burden is only part of the picture. A more pressing issue stems from demographic change. While healthcare costs have long been the focus of reform debates, the real bottleneck of the future is likely to be workforce capacity.

As seen in the graph below, the age distribution of the Dutch population has shifted significantly over time. In 1950, 60.2% of the population was of working age (18–67), while just 5.7%

was 67 or older. By 2005, the working-age group had increased slightly to 66.5%, with the elderly population doubling to 11.5%. In 2025, projections show 64.6% working age and 17.1% aged 67 or older. Looking ahead to 2070, the working-age share is expected to fall further to 59.5%, while the proportion of people aged 67 and above will rise to 22.7%.



Source: CBS

This rising dependency ratio means fewer working people supporting a growing population of elderly citizens, many of whom will need increasingly complex care. If current trends continue, a disproportionate share of the workforce would be required in healthcare simply to sustain current service levels. This is not financially, politically, or socially sustainable.

To better understand this evolution, it is helpful to apply the lens of David M. Cutler (2002), who described healthcare reforms as evolving in three broad phases. These phases are clearly visible in the Dutch context (see e.g. Schut (2003)).

1. Access Expansion (roughly 1945–1980 in the Netherlands): In the decades following the Second World War, the main focus was to increase access to healthcare. Hospitals were expanded, insurance coverage was broadened, and the foundations of the modern welfare state were established. The priority was equity: ensuring that everyone could receive care when needed.

2. Cost Containment (roughly 1980–2000): Rising healthcare expenditures led to a shift in focus. The oil shocks and financial crises of the late 1970s and early 1980s led to budgetary concerns. Policymakers introduced price regulation, budget ceilings, and efficiency incentives. In this phase, a predecessor of my former employer, the Dutch Healthcare Authority, was created to regulate the health care sector. The measures were effective

in curbing spending growth, but came at a cost: waiting lists lengthened, and public satisfaction declined.

3. Incentives and Market Mechanisms (2000s onward): In response to the shortcomings of the cost-containment phase, a third wave of reform started in 2005: the Netherlands introduced regulated competition, insurer choice, and performance-based payment mechanisms. The goal was to improve process efficiency, in other words, to do things right.

Each phase reflected the priorities of its time: from equity to sustainability to efficiency.

But it is worth asking: was it a coincidence that we introduced competition and choice-based systems in 2005, just when the ratio of working-age people to older citizens was at its most favorable? Labor was still abundant, and a model based on patient choice and provider competition was feasible.

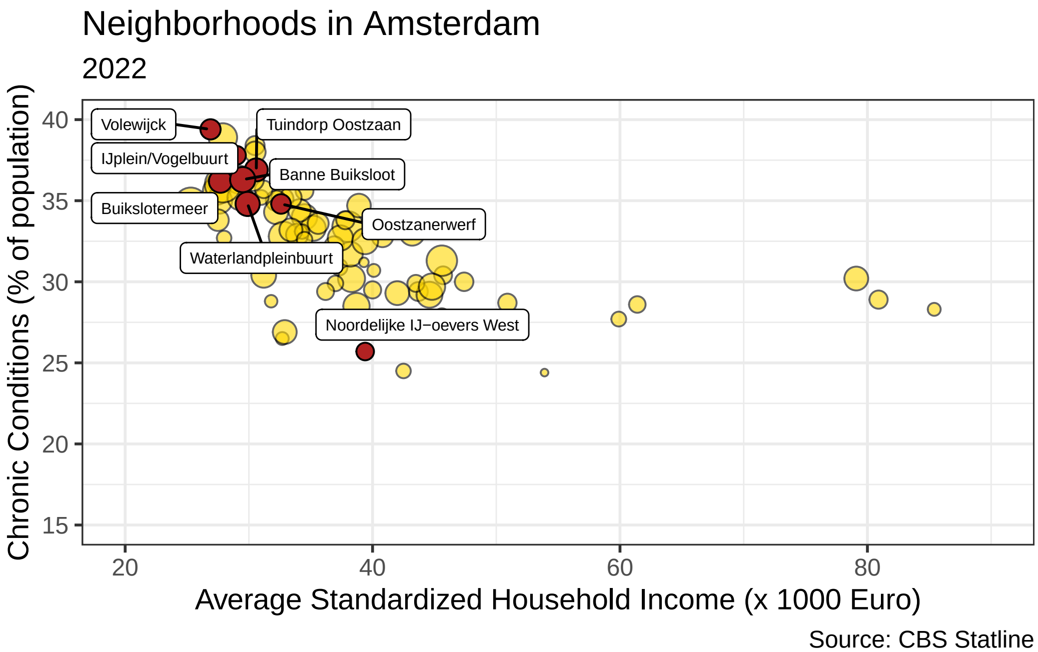
Today, that context is changing rapidly. The real question going forward is not just how (cost) efficiently we can deliver care, but what care we deliver, where, and to whom. The focus is shifting from process efficiency to allocative efficiency: from doing things right to doing the right things.

There is good reason to believe we are not allocating our healthcare resources to the right places. As the OECD (2023) puts it:

> "Unhealthy lifestyles and poor environments cause millions of people to die prematurely. Smoking, harmful alcohol use, physical inactivity and obesity are the root cause of many chronic conditions."

In other words, many of our most pressing health problems do not originate in hospitals or GP practices, but in daily life. This point is illustrated in new research using Dutch administrative data. Danesh et al. (2024) show that health inequality in the Netherlands begins early: by age 35, individuals in the bottom half of the income distribution already carry the same chronic disease burden as 50-year-olds in the top half. This is not just the result of reverse causality (i.e., sicker people earning less), but primarily due to faster health deterioration among low-income individuals. The researchers conclude that socioeconomic and geographic factors, not individual behavior alone, account for most of the difference.

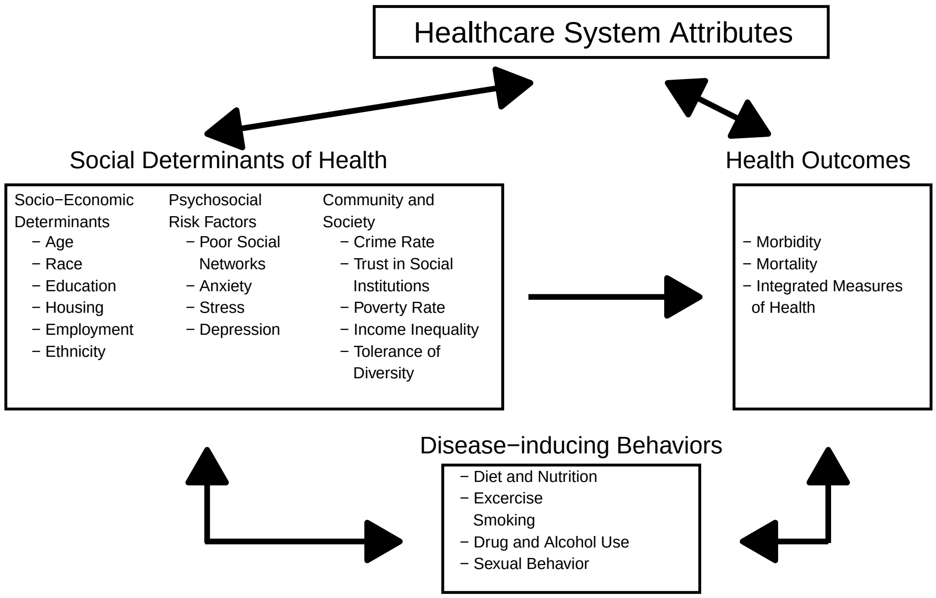
This pattern is also visible at the local level. The figure below plots average income against the prevalence of chronic conditions for neighborhoods in Amsterdam. It shows a clear picture: wealthier areas tend to have lower rates of chronic disease. Neighborhoods in Amsterdam-Noord, highlighted in red, illustrate how certain areas consistently perform worse across both dimensions: suggesting social disparities that cannot be explained by healthcare access alone.



Initiatives such as Beter Samen in Noord seek to address this challenge in practice. By initiating collaboration between healthcare providers, social workers, and community organizations, they aim to reduce health inequalities by intervening earlier and closer to the root causes. This means supporting people not only when they are sick, but also by shaping the environments in which they live, work, and grow up.

A useful visual model based on Ansari et al. (2003) -shown below- illustrates this multi-level framework[1]. It highlights how health outcomes are shaped by a wide range of factors, including socioeconomic position, psychosocial stressors, and broader community characteristics, not just individual behavior or the quality of healthcare services.

[1]Over the last decades, a growing amount of research suggests that medical care is not the sole determinant of health outcomes. Rather, social determinants of health (SDH) play a large role in shaping health outcomes (Braveman and Gottlieb 2014). The WHO defines SDH as “the conditions in which people are born, grow, live, work, and age, and the fundamental drivers of these conditions” (Marmot et al. 2008). This summary is based on research assistance by Finn Mikkers.



This text presents a framework for rethinking healthcare under these conditions. It builds on the field of health systems engineering, which views healthcare as a complex, decentralized system of interdependent actors. In the chapters that follow, I explore the need for improved coordination (Chapter 2), better-aligned payment schemes (Chapter 3), and the consequences for our Dutch healthcare system (Chapter 4).

2 Coordination: Doing the Right Things

2.1 Introduction

In the previous chapter, I argued that the central challenge facing modern healthcare systems is not only financial, but structural: a growing mismatch between the rising demand for care and the limited supply of qualified professionals. This bottleneck calls for more than optimizing existing processes, we need to rethink what care is delivered, by whom, and how it is organized. In short, we must shift our focus from technical efficiency (doing things right) to allocative efficiency (doing the right things).

Too often, health system reforms have concentrated on making individual organizations or care processes more efficient in isolation, what we might call subsystem optimization. Hospitals are streamlined, GP practices digitalized, and care pathways standardized, but these efforts frequently fail to deliver better system-wide outcomes. As Berwick, Nolan, and Whittington (2008) argue in their formulation of the Triple Aim, improving population health and reducing costs requires more than improving individual components, it demands coordinated action across the whole system. Similar concerns are raised by Valentijn et al. (2013), who emphasize that the integrative function of care, especially primary care, is often overlooked when efficiency is pursued in silos.

This fragmentation is not unique to any one country. The OECD (2023) points out that many countries struggle with disjointed care delivery, even when individual providers or sectors function well. Fragmentation persists because optimization at the micro-level, of specific organizations or sectors, does not automatically translate into macro-level system performance.

But today's challenges, rising demand, increasing complexity, and especially labor scarcity, make this kind of fragmented optimization unsound. We cannot improve the system by tweaking existing processes. Instead, we need a shift towards allocative efficiency: doing the right things, at the right place, with the right people.

This shift requires better coordination: the deliberate alignment of actors, incentives, and information across sectors and organizations, also beyond the healthcare sector in the traditional sense. Better coordination helps us reallocate care more sensibly, prevent duplication, and improve outcomes.

The following sections develop this line of reasoning. I argue that in times of labor scarcity and rising demand, coordinated, system-level thinking must replace fragmented subsystem optimization.

2.2 Barriers to Coordination

Efforts to improve healthcare often focus on optimizing discrete components of the system, specific institutions, specialties, or processes, without addressing how these elements interact. As discussed in the previous section, this approach fails to achieve allocative efficiency and often results in fragmented care, inefficiencies, and unmet patient needs. Below, I outline four systemic barriers that hinder better coordination across healthcare systems.

2.2.1 Misaligned Incentives

One of the most persistent barriers to coordination is how healthcare is financed. Budgets are typically divided between medical, social, preventive, and long-term care services, each governed by different rules, actors, and incentives. Hospitals are paid for treatments, not for avoiding admissions. GPs are paid per consultation, not for coordinating care. Municipalities fund social care, while health insurers fund curative care, often without aligned objectives.

This setup works against prevention and integrated care. For instance, as Danesh et al. (2024) show, people with lower incomes in the Netherlands develop chronic conditions like diabetes much earlier in life. We treat everyone equally once they become patients, but too late. The system provides equal treatment but not equal opportunity for health, resulting in unnecessary suffering and avoidable use of scarce healthcare labor.

These coordination problems become even more complex when not just health insurers, but also municipalities and long-term care purchasing agencies (‘zorgkantoren’) are involved. Each of these actors operates under different legal frameworks and financing structures, which can create incompatible incentives across patient pathways. For instance, municipalities may prioritize short-term social care budgets, while insurers focus on medical care costs, leading to mismatches or inefficiencies. Addressing such cross-sectoral misalignment requires careful contractual design and potentially national-level alignment on goals and responsibilities.

I return to these financing issues in detail in the next chapter, where I explore how payment models can be redesigned to support better coordination and long-term health outcomes.

2.2.2 Institutional and Disease-Centered Fragmentation

Healthcare systems remain largely organized around institutions, hospitals, general practices, and home care agencies, and around individual diseases. Clinical guidelines and care protocols often follow a single-disease model, even though multimorbidity is now the norm, especially among older adults, see e.g. Boyd et al. (2005). This disease-centered logic fragments care pathways and ignores how patients actually experience illness, typically as a set of interconnected physical, mental, and social challenges. These single disease focused guidelines may be harmful for patients (Care of Older Adults with Multimorbidity 2012).

Patients with multiple chronic conditions may interact with a range of providers across different locations, each using separate IT systems, operating under different payment models, and pursuing distinct treatment goals. Transitions of care, such as from hospital to home, or between physical and mental healthcare, are often poorly coordinated. This leads to duplication of diagnostics, inconsistent medication prescription, and confusion for patients and families, see e.g. Schoen et al. (2005) and Nolte and McKee (2008).

2.2.3 Information Gaps and Poor Data Sharing

Even when providers want to coordinate, they are often blocked by fragmented information systems. Electronic health records (EHRs) are typically siloed across institutions, and interoperability remains limited (OECD 2019). Legal, technical, and organizational barriers delay or prevent timely data exchange, particularly across sectors such as medical and social care.

A common consequence is the so-called "wrong bed day" (delayed hospital discharge or bed blocking) phenomenon: patients remain in hospitals longer than medically necessary, not because they still require acute care, but because appropriate follow-up care, such as in a revalidation facility, short term residential care (ELV in Dutch), or district nursing care, cannot be arranged in time. These delays are not only costly but also compromise patient recovery and reduce hospital throughput, see e.g. Kripalani et al. (2007) .

2.2.4 Cultural and Behavioral Barriers

Beyond institutions and data systems, fragmentation also exists at the cultural and professional level. Different sectors: medical, social, and mental health, operate with distinct routines, vocabularies, and professional norms. General practitioners, hospital specialists, district nurses, social workers, and mental health professionals may all be committed to patient care, but they often differ on priorities, responsibilities, and interpretations of what good care entails.

These professional boundaries can obstruct cooperation. As shown in a systematic review (Schot, Tummers, and Noordegraaf 2020), professionals often work to bridge gaps between roles, negotiate responsibilities, and create collaborative spaces. However, such efforts are not always structurally supported. Without intentional support for interprofessional collaboration, team-based care can become fragmented.

Professional autonomy, while essential for trust and motivation, can become a barrier when cross-sector collaboration is required. Teams that function well within one organization may struggle to work across settings. As Edwin van der Meer (BovenIJ Hospital, Beter Samen in Noord) noted in an interview embedded in the inaugural adress, "knowing each other", both personally and professionally, is a critical success factor for effective coordination. While the success of Beter Samen in Noord highlights the potential of trust-based collaboration, it also underscores the fragility of such arrangements in the absence of formal support structures or aligned financial incentives.

Without mutual understanding and trust, even well-intentioned partnerships risk remaining superficial. Sustained collaboration demands more than organizational alignment; it requires a culture of shared responsibility and the development of interpersonal relationships across sectoral boundaries.

2.3 Our approach: Health Systems Engineering

2.3.1 Introduction

The previous section described the persistent barriers to effective coordination in healthcare. These are not isolated problems, they are systemic. Addressing them requires a shift in how we understand and organize healthcare. I propose an approach rooted in Health Systems Engineering (HSE): a blend of economic contract theory, systems thinking and systems engineering.

Our approach to Health Systems Engineering (HSE) offers a conceptual framework for redesigning healthcare. It views healthcare as a complex, interdependent system in which improving overall performance depends on optimizing the interactions between actors, not just improving isolated units. The goal is to enable better coordination across the system.

Coordination, however, does not emerge automatically. It must be designed and sustained through incentives that align the goals of individual actors with broader system objectives. Motivation is essential for coordination: actors need to be supported -not punished- for doing the right things, even if those actions do not benefit their own silo under current rules.

At the same time, the system must minimize transaction costs, support dynamic adaptability, and respect professional autonomy. Our approach to HSE draws from contract theory (Bogetoft and Olesen (2004) and Bogetoft and Olesen (2007)) to structure incentives and decision rights, and from systems thinking (Sterman (2000), Bonnema (2023)) to handle complexity, feedback, and unintended consequences.

2.3.2 Systems Perspective: Layers, Interfaces, and Subsystems

I begin with the systems perspective. Healthcare is not a single system, but a set of interacting subsystems. For example, an Emergency Care Department is a subsystem within a hospital; the hospital is part of a larger healthcare system alongside general practitioners, district nursing, mental health providers, and other institutions. This healthcare system, in turn, interacts with other supersystems, such as education, housing, labor markets, and social security. Together, these constitute a broader social system -a supersystem- in which health is both an outcome and a determinant.

Each of these systems is layered. I distinguish three levels:

1. Cultural elements: trust, leadership, professional identity, shared values;
2. Organizational structures: roles, governance, networks, institutions;
3. Financing mechanisms: budgets, incentives, contracts, accountability.

Problems often emerge at the interfaces, boundaries between subsystems (e.g., hospital and home care) or between systems (e.g., medical and social care). For instance, "wrong bed days" arise when a patient no longer needs hospital care but cannot transition to appropriate follow-up care due to organizational or financial misalignment.

Rather than focusing only on improving individual components, HSE emphasizes the interdependencies among them. Optimizing one part of the system, without attention to its interactions, can degrade performance elsewhere. Our framework combines contract theory and systems thinking to guide the redesign of those interactions: aligning incentives, allocating decision rights, and enabling mutual adjustment across settings. In the following chapter, I focus on payment models not because they are the only important element for change, but because they expose the broader structural challenges of coordination, motivation, and accountability in fragmented systems. They offer a concrete entry into more holistic system redesign.

2.3.3 Contracts, Coordination, and Motivation

Building on the insights of Bogetoft and Olesen (2004), Bogetoft and Olesen (2007) and Milgrom and Roberts (1992), I understand contracts not just as legal documents, but as tools for organizing systems. In this view, contracts are institutional tools that shape behavior, allocate responsibilities, and structure interactions across actors within a system. A well-designed contract performs two essential functions:

1. Coordination: ensuring the right actors take the right actions at the right time and place;
2. Motivation: aligning the incentives of autonomous actors to pursue system-level goals.

These functions are connected. In decentralized healthcare systems, motivation is a prerequisite for coordination. Effective contracts must create incentives not only to deliver services, but to do so in a way that stimulated collaboration, learning, prevention, continuity of care, and the reduction of avoidable care such as hospital admissions. If providers are expected to engage in undertaking coordinated actions such as collaboration, joint planning, information sharing, or investing in transitional care, the financial structure must make such actions worthwhile. Their costs must be recoverable, and their expected benefits must outweigh the opportunity costs. Coordinated action must not be a loss-making activity, otherwise, even when system-wide benefits are evident, the absence of individual incentives will lead to fragmentation, underinvestment, and strategic behavior.

In addition to coordination and motivation, two further aspects are essential. First, transaction costs (such as administrative burden, reporting and lobbying cost) must be minimised, not only in the design and negotiation of contracts, but also in their monitoring and enforcement. Second, contracts must be dynamically adaptable: they must be robust to uncertainty, capable of accommodating change, and supportive of learning over time.

These four elements, coordination, motivation, transaction costs, and dynamic adjustment, form the core concerns of HSE.

The structural and behavioral barriers discussed in this chapter all point to a common need: better coordination across the healthcare system. Yet coordination alone is not enough. The systems we've built reward thinking and acting in silos, fragmented delivery of care, and short-term cost control. To overcome these constraints, we need financial mechanisms that not only support integration but actively incentivize it.

In the next chapter, I zoom in on payment schemes as a key design element. Payment schemes are not just financial mechanisms: they shape behavior, coordination, and motivation. By exploring how payment models support or hinder system goals, I highlight why contract theory and systems thinking are essential tools in building future-ready healthcare systems.

3 Payment Schemes and Incentives

3.1 Introduction: What Should a Good Payment System Achieve?

All healthcare systems seem to face a critical problem: fragmentation of services and misaligned incentives both undermine the coordination of care and the delivery of high-value outcomes. Providers often focus on optimizing isolated subsystems rather than collaborating across the patient pathway. The design of payment systems has been widely recognized as a crucial element in healthcare system reform (see e.g Fainman and Kucukyazici 2020), influencing how resources and activities are allocated, how providers coordinate across patient pathways, and how financial risks are distributed.

A good payment system must do more than simply reimburse services. It must:

- Encourage allocative efficiency: ensuring that resources are used for the right care, by the right provider of the right quality at the right time.
- Align incentives across the patient pathway: stimulate collaboration among different providers. The payment model should ensure that a participant is willing to be involved in a collaboration (which means that participation should not be loss making) and that the collaboration chooses actions that align with the overall desired outcome.
- Reward prevention and outcomes, rather than volume of services.
- Be implementable and adaptable to changes and real-world complexity.

To achieve these goals, we should carefully design payment models that influence provider behavior in ways that promote coordination, improve outcomes, control costs and avoid inefficient resource use. However, we should not expect to achieve a first-best optimum. As Arrow (1963) famously argued, healthcare markets are subject to market failures, including information asymmetries, uncertainty, and externalities, that make perfect outcomes unfeasible. Instead, the aim is to design payment systems that move us closer to better outcomes within the constraints imposed by these imperfections.

In this chapter, I answer four key questions that together guide the design of better payment schemes:

1. What should we pay for?
2. How should we pay?

3. Who will be paid?
4. Who pays?

To improve coordination among healthcare providers and move beyond the optimization of isolated subsystems, we need to focus on how to align incentives within a framework of decentralized decision-making. This requires a careful examination of payment schemes.

First, we must consider what to pay for. This involves defining the scope of services and patient pathways included in payment models. Moving away from fee-for-service models towards payment models for broader bundles of activities, population-based payments may encourage providers to focus on entire patient journeys rather than isolated treatments (see e.g. Bogetoft, Mikkers, and Shestalova (2019)).

Second, we need to determine how to pay. Exploring different payment methods, such as global budgets, capitation, or bundled payments, and integrating quality metrics into these schemes can help align provider incentives with desired health outcomes. This ensures that providers are rewarded not just for the quantity of services delivered but for the quality and effectiveness of care.

Third, we should take into account the impact of payment models on organization and governance. Understanding how these financial models influence the organizational structures and governance mechanisms among healthcare providers is important. Better payment schemes can stimulate collaboration, reduce fragmentation, and give incentives for coordinated care delivery by encouraging providers to work together toward common goals.

Fourth, we must consider the organization on the funder side. This involves examining the roles and interactions of governments, insurers, and municipalities as payers, and how their contracting relationships shape the funding and delivery of healthcare services. Clarifying these relationships can help streamline funding flows and reduce administrative complexities that hinder coordination.

Through this analysis of payment schemes, I arrive at the conclusion that implementing regional budgets may be an effective way to enhance coordination and efficiency in the healthcare system. By allocating budgets based on predicted healthcare costs for specific populations, regional entities can be incentivized to tailor services to the needs of their communities. However, I should acknowledge that we have not yet fully resolved the complexities related to organization, governance, and the roles of funders in this model. These aspects are disccused further in Chapter 4.

To stimulate continuous learning, improvement and the adoption of best practices, I propose utilizing yardstick competition. By benchmarking regions against one another, we can drive efficiency and quality improvements across the system. This method motivates regions to learn from each other, supporting a culture of innovation and collaboration that ultimately improves overall healthcare performance.

3.2 What to pay for?

In designing more effective payment schemes, it is essential to determine what exactly we are paying for. Healthcare is not delivered in isolated events but as a series of interdependent activities along a patient pathway, from prevention to diagnosis, treatment, rehabilitation, and long-term follow-up. Poorly aligned payments fragment these pathways and undermine both quality and efficiency (Gaynor, Ho, and Town 2015).

Traditional fee-for-service (FFS) models compensate individual services separately and therefore, tend to stimulate fragmentation. They reward volume rather than value and discourage coordination between providers (Porter, Kaplan, et al. 2016).

To overcome these inefficiencies, payment models should cover entire care episodes or population-based pathways. For example, an elderly patient with osteoarthritis may require coordinated care: lifestyle interventions, medication management, surgery, post-surgical rehabilitation, and chronic care monitoring. Paying separately for each service risks duplication, gaps, and cost escalation. Instead, bundled payments for the entire episode gives incentives to providers to collaborate and optimize the over full patient path.

Moreover, healthcare pathways involve complementary services, where interventions like physical therapy may enhance surgical recovery, and substitutable services, such as lifestyle changes replacing medication in early disease management. Payment models should recognize these relationships to avoid inefficient resource use and improve patient outcomes (see Agrell, Bogetoft, and Mikkers (2013) in another setting).

Effective payment design must also account for externalities, the broader system-wide effects of individual interventions. Investment in preventive services, for example, reduces future hospitalizations and improves long-term health, but the benefits often accrue to different parts of the system (see e.g. Bogetoft, Mikkers, and Shestalova (2019)).

Addressing these challenges requires payment models that align incentives across the patient pathway, encouraging providers to coordinate services and focus on achieving the best outcomes over the full cycle of care (Berwick, Nolan, and Whittington 2008).

3.3 How to pay?

The way healthcare services are paid for has effects on provider behavior. Payment schemes not only determine the level of care provided but also shape incentives for quality, coordination, and innovation. Moving beyond simple price-per-service models requires careful contract design to balance incentives and risks. Healthcare payment models fundamentally shape provider behavior by balancing incentives and risk. A payment model that offers strong incentives to improve quality and efficiency may also expose providers to financial risks they cannot control. Conversely, models that limit risk for providers often weaken their motivation to innovate or optimize care delivery (see e.g. Bogetoft and Olesen (2004) and Laffont and Martimort (2009)).

Regardless of the model chosen, quality metrics must be integrated into the payment system from the beginning. Monitoring outcomes, such as real health outcomes or intermediate outcomes such as patient recovery rates, readmission rates, and patient-reported experiences, is crucial to prevent cost control from degenerating into underprovision or inappropriate care (Fainman and Kucukyazici 2020).

Economic theory shows that there is no ideal solution. Instead, payment systems must find a practical balance: enough incentive to drive performance, but not so much risk that providers engage in undesirable behaviors such as cherry-picking or the under production of necessary care (see Bogetoft and Olesen (2004) and Laffont and Martimort (2009)). As healthcare delivery becomes more complex and more integrated, payment models must evolve along: from fragmented payments with low provider risk to more integrated models with higher provider responsibility.

I review some standard payment models along this continuum.

3.3.1 Fee-for-Service (FFS) and In-Hospital Bundles (DRGs, DBCs, HRGs)

The traditional fee-for-service model reimburses providers for each service delivered. While easy to administer, it encourages providers to maximize the number of services, leading to overproduction and fragmentation of care. Pure fee-for-service is rare today; most hospital systems use some form of bundling, such as Diagnosis-Related Groups (DRGs) in the United States, DBC (DTC) in the Netherlands, or Health Related Groups (HRG's) in the United Kingdom. These models group hospital services related to a specific condition but usually remain confined to a single provider.

- Risk: Primarily on the payer.
- Incentives: Strong for volume; little incentive for coordination; fragmentation remains.

3.3.2 Two-Part Tariffs

Two-part tariffs combine a fixed component (e.g., capitation) with a variable, activity-based payment. For example, Dutch general practitioners receive a base capitation payment supplemented by a fee-for-service element (Kroneman, Van der Zee, and Groot 2009). This model balances the need for stable income with incentives for productivity, but its effectiveness depends heavily on how the fixed and variable parts are calibrated. A low variable component may not sufficiently stimulate effort, while a high variable component can encourage overproduction. Proper case-mix adjustment is essential to prevent providers from avoiding high-risk or complex patients.

- Risk: Shared between payer and provider, depending on the mix.
- Incentives: Moderate; adjustable with careful calibration.

3.3.3 Bundled Payments

Bundled payments extend the bundling concept across multiple providers and stages of care, for example, a hip replacement bundle covering preoperative consultations, surgery, rehabilitation, and follow-up care (Struijs et al. 2020). They incentivize coordination across the patient pathway and have shown potential for improving outcomes and reducing costs. However, bundled payments also introduce new complexities. To prevent providers from selectively treating low-risk patients, a phenomenon known as cherry-picking, careful case-mix adjustment is essential. Additionally, bundled payments require clear contractual arrangements among participating providers to define how payments and responsibilities are shared. Without clear agreements, coordination problems and disputes can arise, undermining the model's intended benefits. Finally, success depends on establishing shared accountability for outcomes, ensuring that all providers involved in a patient's care pathway are jointly responsible for both the quality and cost of care.

- Risk: Shared among multiple providers.
- Incentives: Strong for coordination and efficiency; requires careful case-mix adjustment to prevent cherry-picking.

3.3.4 Global Budgets

Global budgets allocate a fixed total budget to cover all healthcare services for a defined population over a set period. By placing financial responsibility on providers, global budgets create strong incentives for cost control and resource prioritization. However, strong financial incentives also transfer significant risk to providers, who may respond by reducing necessary care to remain within budget (see e.g. Bogetoft and Olesen (2004) in other settings). Without effective quality monitoring and case-mix adjustments, there is a risk that providers prioritize cost savings over patient outcomes.

To address these concerns, models such as Accountable Care Organizations (ACOs) in the United States and the Alternative Quality Contract (AQC) introduced by Blue Cross Blue Shield of Massachusetts have incorporated more sophisticated risk-sharing mechanisms with the payers. In these models, providers are eligible for shared savings if they manage to deliver care below predefined cost benchmarks. However, simply rewarding cost savings risks encouraging underprovision. To mitigate this, these contracts use asymmetric profit-sharing arrangements: providers can retain a larger share of savings if they also meet established quality targets, but gain little or nothing if quality falls below threshold standards (Song et al. 2019).

Another prominent example is Gesundes Kinzigtal in Germany. This model operates under a population-based global budget, where healthcare providers are financially responsible for the total cost of care for a defined population. Providers are incentivized to improve health outcomes and reduce avoidable healthcare use, and they share in any savings achieved. Gesundes Kinzigtal combines global budget principles with strong emphasis on preventive care, patient engagement, and the use of data analytics to monitor performance, offering valuable lessons on the potential and challenges of such payment reforms, see e.g. Bogetoft, Mikkers, and Shestalova (2019) and Larrain and Groene (2023) for a detailed discussion of the Gesundes Kinzigtal model.

Additionally, many global budget models incorporate provisions which limit the financial exposure of providers in cases of exceptionally high-cost patients. These mechanisms ensure that while providers have incentives to deliver efficient care, they are not overly penalized for random variation in patient costs.

- Risk: Predominantly borne by providers, but shared with payers through shared savings.
- Incentives: Strong incentives for both cost control and quality improvement, provided that quality metrics are closely tied to financial rewards.

3.4 Who to Pay? Organizing Providers

To align incentives along the patient pathway, we not only need to ask how to pay, but also who should be paid. Payment models influence provider behavior, but the effectiveness of these incentives depends on how care is organized and how responsibilities are distributed among providers. Fragmented organizational structures often undermine coordination, even when financial incentives are well-designed. Therefore, designing effective payment systems requires a parallel discussion about governance and contractual forms among providers.

Effective provider collaboration goes beyond financial contracts, it requires trust, shared accountability, and aligned goals. As Aunger, Millar, and Greenhalgh (2021) argue, successful inter-organizational collaboration in healthcare depends on the interplay of trust, confidence, and faith between actors. These elements are not only shaped by institutional arrangements but also by histories of cooperation, expectations of behavior, and mutual understanding. Especially when care spans different sectors, such as medical, mental health, and social care providers must be able to make joint investments and resolve conflicts over funding, responsibilities, and priorities. That calls for new governance models not just within organizations, but across institutional boundaries.

Three broad governance forms are commonly discussed in the literature and practice.

3.4.1 Main Contractor Models

In this structure, one lead organization, often a hospital, holds the main budget and subcontracts other providers. While this can streamline financial flows, centralize responsibility and limit informational asymmetries (Agrell, Bogetoft, and Mikkers 2013 in a different setting), it also introduces hierarchies that may limit professional autonomy and undermine genuine cooperation. The main contractor may lack incentives to invest in coordination or prevention if those benefits accrue to subcontractors. Dutch experiments with this model, such as the "keten-DBCs" in diabetes care, have shown mixed results (Kroneman, Van der Zee, and Groot (2009); Struijs et al. (2021)).

3.4.2 Cooperative Models

An alternative to hierarchical contracting is the cooperative model, where multiple providers jointly hold the contract and share responsibility for spending, outcomes, and governance. This model emphasizes horizontal collaboration, with shared decision-making, distributed risk, and collective accountability.

Cooperatives might be promising when collaboration must occur across sectoral or institutional boundaries, for example, between medical care, mental health, and social services. They create incentives for mutual support and coordinated investment in prevention or care transitions that benefit the system as a whole.

As Bogetoft and Olesen (2007) explain in the context of Danish agriculture, cooperatives succeed when internal payment schemes and decision rules align with individual members' incentives. That means each participating provider, a hospital, a GP group, or a social care organization must be better off within the cooperative than outside of it. Importantly, no subgroup (a sub-coalition) should have an incentive to break away, which requires careful contract design and fair surplus distribution.

These insights are crucial in healthcare, where provider types differ in size, cost structures, and patient populations. For cooperatives to function, internal contracts must acknowledge these differences while maintaining collective efficiency. This is often challenging in practice, as governance rules must avoid dominance by larger or more powerful members, and outcomes must be jointly attributable and measurable.

In my own experience with regional cooperation in the Netherlands, I find that many healthcare providers are willing to collaborate in principle but hesitate when it comes to financial interdependence. Cooperative models require a level of transparency, mutual trust, and long-term commitment that is still rare in many healthcare markets. Dutch experiments such as Beter Samen in Noord and De Zeeuwse Zorgcoalitie look promising but also revealed insurers' reluctance to engage with cooperatives, often preferring bilateral contracts with individual providers.

Nevertheless, cooperative governance remains a promising way to align incentives across organizations while preserving professional autonomy and encouraging peer-based accountability.

3.4.3 Integrated Delivery Systems

A more centralized model is the integrated delivery system, where providers are merged into a single organization that delivers a wide range of services. Historically, the Health Maintenance Organization (HMO) model was the most prominent form. HMOs combined capitated payment with a closed network of providers who assumed full responsibility for patient care. By integrating financing and delivery, HMOs aimed to control costs and reduce unnecessary care.

However, HMOs faced significant backlash in the U.S. during the 1990s, as critics argued they restricted access, limited patient choice, and incentivized under-provision. This experience shaped the evolution of Accountable Care Organizations (ACOs) in the 2000s (Dugan 2015).

ACOs represent a more flexible, incentive-based evolution of HMOs. Rather than pre-paying providers, ACOs operate on shared savings contracts: if providers can deliver care below a set benchmark while maintaining quality, they share the savings. Conversely, if spending exceeds the target, they may face shared losses. Most ACOs continue to use fee-for-service as a base, but overlay this with performance incentives and retrospective reconciliation.

As Song et al. (2019) demonstrate, long-term experience with global budgets in ACO-like models, such as the Massachusetts Alternative Quality Contract, shows that savings can be sustained if quality incentives are carefully embedded. However, data infrastructure, integration of care, and mutual trust among providers remain prerequisites for success.

A similar integrated care model has been implemented in Germany through Gesundes Kinzigtal, which combines global budgets and shared savings with strong community engagement. Providers and insurers collaborate in a joint venture to improve population health and reduce unnecessary care, supported by real-time data analytics and performance feedback. Evaluations of the program indicate sustained improvements in outcomes and reductions in costs (Groene and Hildebrandt (2025)).

These examples illustrate that joint accountability is critical to avoid cost-shifting and to ensure a shared focus on patient outcomes. Shared savings and risk-sharing mechanisms must be carefully calibrated to encourage collaboration without punishing providers for unavoidable complexity (Berwick, Nolan, and Whittington 2008).

3.5 Who pays? Organizing Payers

In theory, multiple insurers competing under managed competition should drive efficiency and innovation. However, in practice, payer-side fragmentation introduces coordination failures. A central issue is that payment reform by one insurer can generate positive externalities -such as better outcomes, lower hospital use, or spillover learning effects- that benefit all insurers in the region. This creates a public goods problem: no individual insurer has sufficient incentive to invest in innovation alone, since others free-ride on the results.

Our forthcoming work (Mikkers, Berden, and Shestalova 2025) provides a formal model of these dynamics. We show that even when follow-up contracts to a successful pilot (e.g. bundled payments) are socially beneficial and legally permissible, they may fail to materialize due to misaligned incentives between insurers. This inefficiency is not a matter of regulatory restriction, but of strategic behavior in a decentralized system.

To resolve this, follow-up policies are needed, mechanisms that encourage or require joint contracting or collective action when payment innovations prove effective. Such arrangements are increasingly allowed under competition law frameworks, for example, the Dutch Competition Authority permits joint purchasing when it supports "appropriate care." However, permissibility alone may be insufficient: without proper incentive alignment or coordination mechanisms, effective payment models may remain isolated pilots rather than becoming structural improvements. For example, ParkinsonNet[1], despite its proven cost-effectiveness and quality improvements, still depends largely on temporary innovation grants (Bloem et al. 2017).

Furthermore, these challenges are even harder when different types of payers, such as health insurers, long-term care purchasers, and municipalities, must collaborate. Each operates under a distinct legal regime and serves overlapping populations. As a result, the need for coordination is both more urgent and more difficult, reinforcing the call for institutional frameworks that enable shared responsibility and long-term investment.

[1] ParkingonNet is a Dutch healthcare initiative that organizes care for people with Parkinson's disease through regional networks of different types of providers.

4 Towards Integrated Health Systems: Scenarios and Strategies

4.1 Introduction: System-Level Consequences

Improving healthcare delivery requires more than refining payment models in isolation. As outlined in the previous chapters, many of the systemic problems we face (fragmentation, misaligned incentives and poor coordination), stem not from payment alone, but from deeper organizational and institutional structures. A reform agenda must therefore address three interconnected layers of the healthcare system:

- Culture: stimulate mutual trust, shared responsibility, professional identity, and leadership. Coordination begins with a culture that values collaboration across boundaries. Leadership at all levels is needed to bridge professional silos, set shared goals, and support change. A data-driven approach to learning is crucial to strengthen professional autonomy and continuous improvement.
- Organization: enable trust-based cooperation through governance, shared rules, and institutional support. Cross-sector collaboration requires more than goodwill. It demands clear rules for joint decision-making (e.g., on shared investments or workforce planning), mechanisms for resolving conflicts, and governance arrangements that lower transaction costs and align roles and responsibilities across institutions.
- Financing: align payment models with patient needs and long-term system goals. Payment is not neutral, it influences what care is delivered, by whom, and how. Financial mechanisms must reward coordination, continuity, and prevention. They must also be adaptable to local contexts, account for complexity, and distribute resources in ways that enable integrated, person-centered care rather than reinforcing fragmentation.

Yet greater coordination might bring unintended consequences. On the provider side, regional collaborations (e.g., Beter Samen in Noord) may become so essential to care delivery that they acquire market power, so they can act independently of insurers. On the payer side, closer coordination between insurers -while necessary to overcome the public goods problem in payment innovation- reduces the remaining mechanism of competition. Since Dutch insurers cannot engage in premium differentiation or risk selection, their only remaining margin for competition lies in better contracting than their competitors. Follow-up contracting (as described in Chapter 3) for successful innovations, while socially desirable, undermines this mechanism.

These dynamics suggest that more vertical integration beteen providers and between payers and providers may be inevitable. Indeed, this is the direction proposed by Ven and Schut (2024), who advocate for competing regional HMO-style systems. My view is that while more vertical collaboration and integration may be the right path, true competition between regional HMOs is unlikely to emerge. Even in dense areas like Amsterdam-Noord, the integrated network of Beter Samen in Noord leads to a (near) regional monopoly.

Thus, any realistic scenario for the future must assess these effects and weigh trade-offs. This chapter does not offer predictions, but explores three scenarios to illustrate possible trajectories. In each, key questions emerge:

- How large should these regional systems be?
- Who should define their governance?
- What happens to specialized care organized nationally?

These key questions are not yet answered.

4.2 Different Scenario's

4.2.1 Scenario 1: Controlled Integration

In this scenario, regional provider collaborations grow organically, supported by light regulation. Providers integrate services across care domains (hospital, primary, social, mental health), and insurers form voluntary coalitions to coordinate follow-up contracts. Governance structures remain mostly private, with minimal state involvement.

- Pros: Maintains flexibility, encourages local innovation, minimizes bureaucracy.
- Cons: Without strong oversight, coordination may fail.

Providers may gain market power without accountability. Payers may lose incentives to contract for value.

The U.S. DSRIP programs suggest the feasibility of voluntary regional coordination (Janus 2019). However, these initiatives often lack structural follow-up, and their population coverage may remain partial. The central risk is that while the system formally allows for coordination, incentives and market structures discourage sustainable implementation.

4.2.2 Scenario 2: Hard Integration

In this scenario, the system embraces full regionalization. Regional budgets are assigned to integrated care organizations, which assume responsibility for a defined population. The government sets the rules for governance, price and budget regulation, data-sharing, and quality oversight.

- Pros: Strong alignment of incentives, public accountability, potential for cost control and improved outcomes.
- Cons: Bureaucratic rigidity, potential for innovation stifling, difficulty in managing cross-regional or specialized care.

Integrated systems also require high levels of trust, data capacity, and adaptive governance. Without these, hard integration risks becoming a rigid administrative structure that fails to deliver patient-centered care.

4.2.3 Failure to adapt to better coordination

In this scenario, the system remains fragmented. Provider collaborations remain isolated pilots. Insurers continue to contract independently. There is no structural investment in coordination, and innovation depends on temporary grants.

The potential consequences are:

- Growing inequality in access and outcomes.
- Workforce shortages may lead to waiting lists and care delivery challenges.
- Public distrust of the healthcare system.

This is the "default path" if action is not taken. It reflects the trajectory of health systems that acknowledge systemic problems but fail to implement coherent solutions. Ultimately, this scenario leads to crisis-driven reform rather than proactive change.

4.3 Final Reflection: Experimentation and Learning

The scenarios outlined above are not predictions, but illustrations of different pathways, each with different advantages, risks, and trade-offs. What they make clear, however, is that there are no blueprints for system transformation. Structural change in healthcare is complex, uncertain, and politically sensitive. But uncertainty should not be an excuse for inaction.

In such a context, the best strategy is not to wait for perfect designs, but to start small, experiment, monitor, and adapt. Regional pilots, such as Beter Samen in Noord, demonstrate how trust-based collaboration can emerge and deliver results. But their success depends on more than goodwill—it requires institutional support for learning.

This calls for a different way of thinking: one that acknowledges complexity, embraces uncertainty, prioritizes institutional learning over static design and a good analysis of the process of change.

Experimentation in healthcare should not be reduced to isolated, fixed pilots. Real-world interventions often evolve during implementation as stakeholders learn and adjust. This makes experimentation a dynamic process. It requires structured, but flexible efforts to explore what works in practice. Recognizing that outcomes may differ by context, population, and professional culture.

- Adaptive governance is essential to make this work. When no blueprint exists and when outcomes are uncertain, governance must be flexible and resilient. Here, contract theory offers a valuable framework. In particular, the idea of incomplete contracts helps us understand why fixed rules and rigid accountability structures often fail in complex environments. Not all contingencies can be foreseen and not all outcomes can be pre-specified. This means governance should not rely solely on detailed ex-ante contracting but must also stimulate longer-term relationships, mutual trust, and the ability to renegotiate and adapt. In this sense, adaptive governance is the institutional form of contract theory under uncertainty: it is about designing frameworks that evolve as information emerges, as consequences and incentives are better understood through experience.
- Bayesian causal learning offers a good framework for this type of real-time learning and decision making. Because experiments are often adapted along the way, involve small sample sizes, and operate in complex environments, Bayesian multilevel models are a good way to estimate effects under uncertainty. Unlike classical methods that rely on large-sample inference and fixed designs, Bayesian approaches allow us to incorporate prior knowledge, continuously update our beliefs, and quantify the probability of specific effects. This is especially helpful under continous monitoring of the effects. However, in real-world policy settings randomized controlled trials are often infeasible due to ethical, logistical, or political constraints. This makes robust causal modeling essential. Well-designed observational studies using (staggered) difference-in-differences, matching, instrumental variables or regression discontinuity designs can provide causal identification strategies. Bayesian methods can strengthen these approaches by modeling uncertainty explicitly, integrating multiple sources of data, and yielding interpretable probability statements. For example, we can estimate the probability that a particular intervention reduces the incidence of chronic disease by 10% within five years. This information is far more relevant to policymakers than conventional p-values. Bayesian learning also accommodates non-linear and dynamic processes, where average effects may hide important heterogeneity in outcomes, risks, and benefits.

For me, this way of thinking, combining systems thinking, contract theory and Bayesian reasoning, is fundamental to what Health Systems Engineering should be. Health Systems

Engineering is not just about technical optimization, but about designing systems that are capable of learning, adapting, and improving under real-world conditions. It is a practical and principled response to complexity.

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As this lecture has shown, Health Systems Engineering is a complex task, one that I certainly cannot undertake alone. Improving healthcare requires multidisciplinary collaboration, not only between providers, welfare organizations, and payers but also across academic fields.

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